
3.1.6 Mammalian blood possesses a number of defensive functions.

Principles of immunology

Phagocytosis and the role of lysosomes and lysosomal enzymes in the subsequent destruction of ingested pathogens.

Definition of antigen and antibody.

Antibody structure and the formation of an antigen-antibody complex.

The essential difference between humoral and cellular responses as shown by B cells and T cells.

The role of plasma cells and memory cells in producing a secondary response.

The effects of antigenic variability in the influenza virus and other pathogens on immunity.

The use of vaccines to provide protection for individuals and populations against disease.

The use of monoclonal antibodies in enabling the targeting of specific substances and cells.

Candidates should be able to

- evaluate methodology, evidence and data relating to the use of vaccines and monoclonal antibodies
 - discuss ethical issues associated with the use of vaccines and monoclonal antibodies
 - explain the role of the scientific community in validating new knowledge about vaccines and monoclonal antibodies, thus ensuring integrity
 - discuss the ways in which society uses scientific knowledge relating to vaccines and monoclonal antibodies to inform decision-making.
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Biological principles

When they have completed this unit, candidates will be expected to have an understanding of the following principles.

- Proteins and polysaccharides are made up of monomers that are linked by condensation.
- Many of the functions of proteins may be explained in terms of molecular structure and shape.
- Enzymes are proteins and their rates of reaction are influenced by a range of factors: temperature, the presence of inhibitors, pH and substrate concentration.
- Substances are exchanged by passive or active transport across exchange surfaces. The structure of plasma membranes enables control of the passage of substances across exchange surfaces.

The examiners may draw on an understanding of these principles in Units 2 and 3. This understanding may also be required in the A2 Units where it may contribute to the assessment of synoptic skills.

Investigative and practical skills

Opportunities to carry out the necessary practical work should be provided in the context of this unit. Candidates will be assessed on their understanding of *How science works* in all three AS Units. To meet this requirement, candidates are required to

- use their knowledge and understanding to pose scientific questions and define scientific problems
- carry out investigative activities, including appropriate risk management, in a range of contexts
- analyse and interpret data they have collected to provide evidence
- evaluate their methodology, evidence and data, resolving conflicting evidence.

The practical work undertaken should include investigations which consider

- the effect of a specific variable on the rate of reaction of an enzyme-controlled reaction
- the effect of a specific variable on human heart rate or pulse rate
- the effect of solute concentration on the rate of uptake of water by plant tissue.